Visually Grounded Task and Motion Planning for Mobile Manipulation

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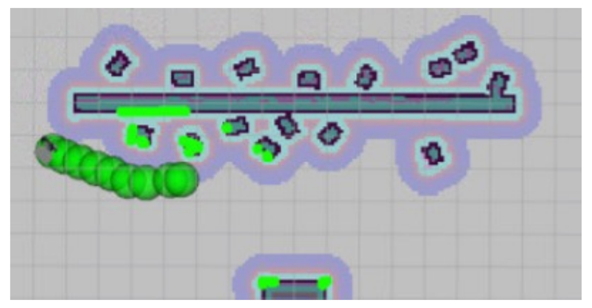
Background

Task planners
- plan in discrete spaces
- sequence symbolic actions
- guide high-level behaviors

% Goal States:
unloadedto(object, location_1).

% Initial States:
at(object, location_0).

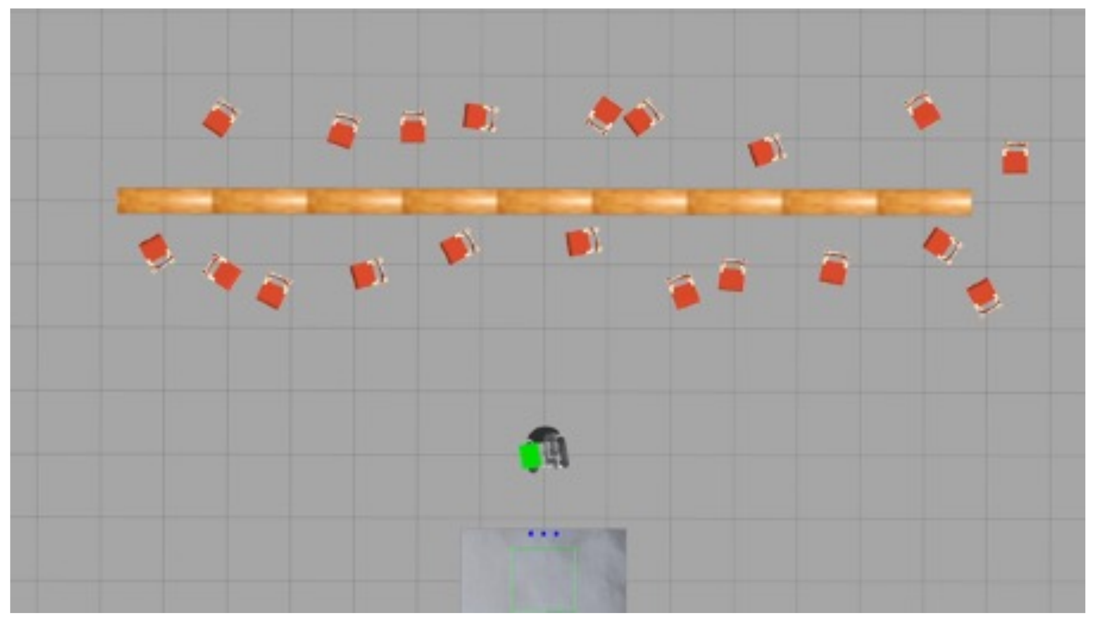
Motion planners
- plan in continuous spaces
- calculate low-level trajectories
- implement symbolic actions



TAMP algorithms aim to bridge the *symbolic-continuous gap* by enabling robots to
- fulfill task-level goals
- maintain motion-level feasibility

Problem

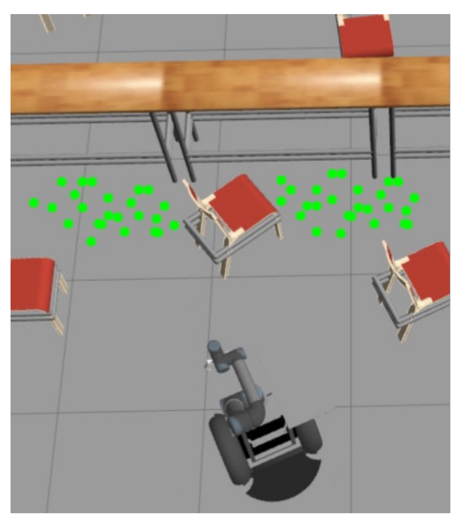
We focus on domains that require robot actions that take relatively long time



The robot needs to navigate to locations from which it can successfully perform the manipulation actions, ideally as quickly as possible.

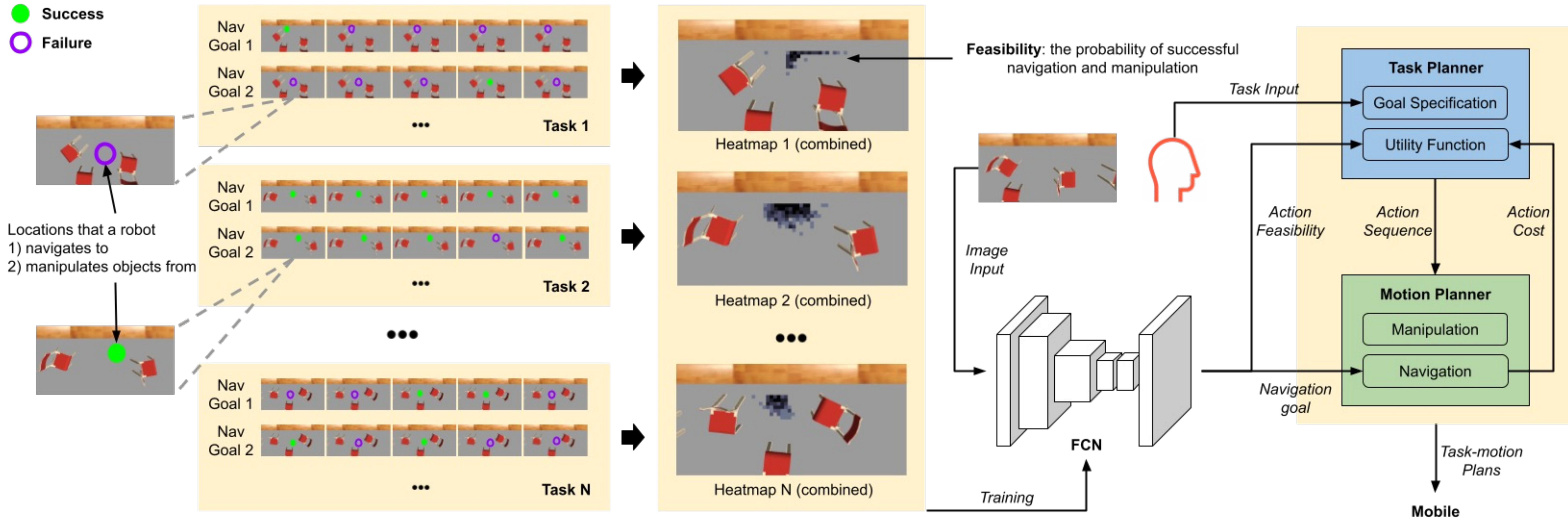
Previous Work

- Deterministic feasibility evaluation
  - IDTMP [Dantam et al., 2018]
  - FFRob [Garrett et al., 2016]
- Predefined state-mapping functions
  - PETLON [Lo et al., 2020]
  - TMP-RL [Jiang et al., 2019]



We introduce **G**rounded **RO**bot Task and Motion **P**lanning (GROP):
- ➢ probabilistically evaluates feasibility
- ➢ learns state mappings from visual perception
- ➢ optimizes both feasibility and efficiency

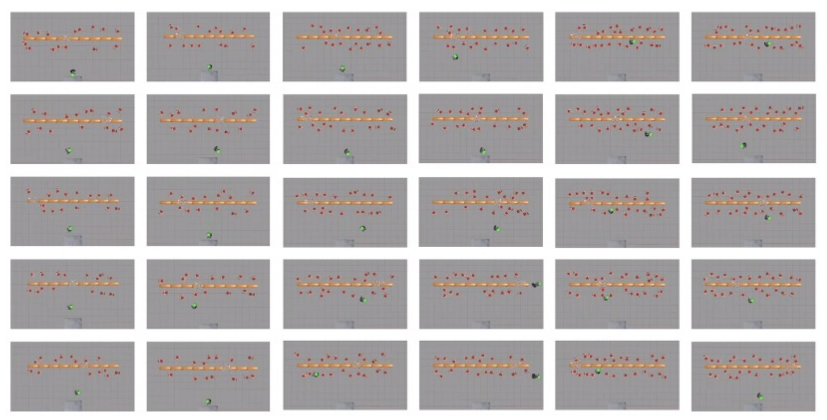
Methods



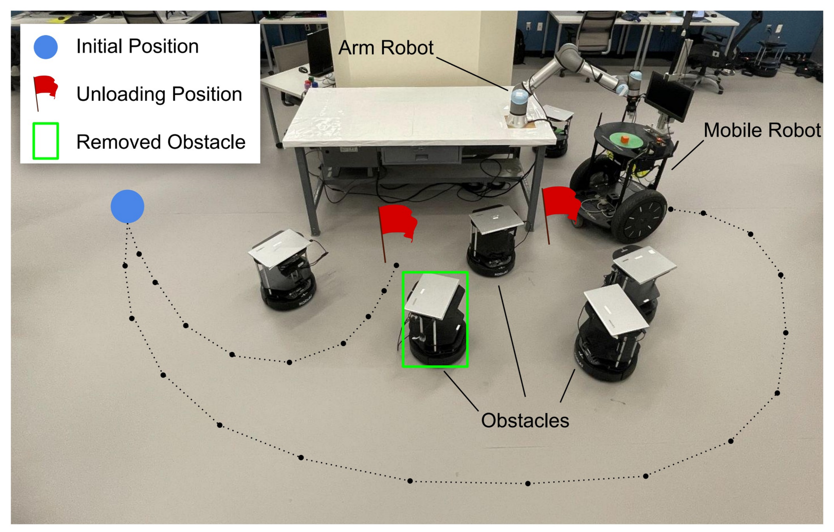
Tasks vary in unloading goals and obstacle configurations.

Dataset

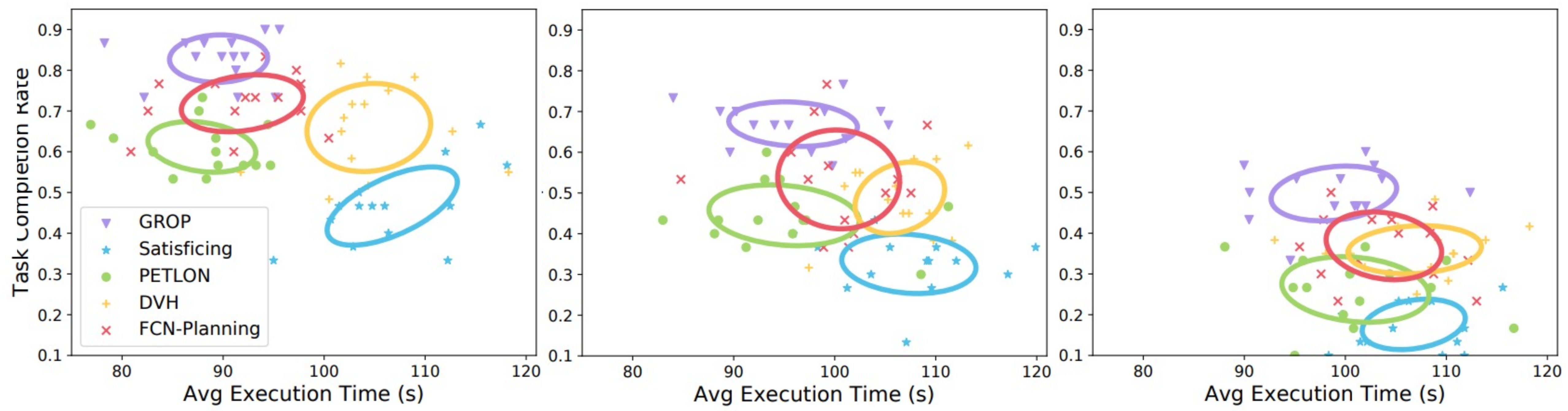
Experiments



Sim



Real



(a) Easy (b) Normal (c) Hard

- ❖ The ellipses represent the means and 2D standard variances
- ❖ Two competitive baselines: PETLON [Lo et al., 2020] and DVH [Driess et al., 2020]
- ❖ **GROP** produced the highest task completion rate while maintaining smaller or comparable execution time.

Conclusion

- ✓ This paper introduces an algorithm, called GROP, that considers both efficiency and feasibility for robot task-motion planning.
- ✓ GROP visually grounds spatial relationships to probabilistically evaluate action feasibility and is particularly suitable for TAMP domains with long-term robot operations (e.g., long-distance navigation).
- ✓ We have extensively evaluated GROP in simulation using a mobile manipulator and demonstrated it using a real robot system that includes a mobile robot and an arm robot.
- ✓ Results showed that GROP outperformed competitive baselines from the literature in plan efficiency without introducing additional action costs.

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